SOUMYA GHORAI

sg22@kgpian.iitkgp.ac.in soumyaghoraijuee2013@gmail.com (+91) 7407875771
linkedin.com/in/soumya-ghorai Google Scholar Personal Website

Education

Indian Institute of Technology Kharagpur, India *Sept 2022 – Present*
Ph.D. Candidate, Electrical Engineering CGPA: 9.5/10
Thesis: Advanced Control and Modulation Techniques for Single-stage DC-linked DAB Converters
Advisor: Souvik Chattopadhyay
Committee Members: Debaprasad Kastha, Santanu Kapat, Pradip Mandal
Recipient of **Prime Minister Research Fellowship 2023**

Jadavpur University, Kolkata, India *2017 – 2019*
M.Tech., Energy Science & Technology CGPA: 8.43/10
Thesis: Sectoral harmonization of electric vehicle transport and power approaching demand side energy management: a case study in West Bengal
Thesis Advisor: Tushar Jash

Jadavpur University, Kolkata, India *2013 – 2017*
B.E., Electrical Engineering CGPA: 7.33/10

Research Interest

My current research focuses on advanced control and modulation techniques for single-stage bidirectional AC-DC and AC-AC dual-active-bridge (DAB)-isolated topologies with boost-type input current shaper and wide-bandgap (WBG) SiC switches. Applications include high-efficiency On-board Chargers (OBCs), solid-state transformers (SSTs), minimum-hardware control-enabled active-power decoupling in OBCs, modular power processing in EV fast chargers, and reduced passive components for AC-AC synchronous soft-open points with reactive-power capability for AC microgrids.

Journal Publications

Nearly Capacitor-less Single-stage DAB-based DC-linked Power Electronic Transformer With Volt-ampere Characteristics of Line Frequency Transformer **TPEL,2026** (Under preparation)
Soumya Ghorai, Souvik Chattopadhyay

Carrier-based Estimated Power Mode Control To Decouple Active Power In A Single-phase Single-stage DAB Converter **TIE, 2026** (Under review)
Soumya Ghorai, Souvik Chattopadhyay

Combined PWM and Phase Shift Control of a Variable DC-Link PFC-Integrated DAB Topology for Wide-Range High-Efficiency EV Chargers **JESTPE, 2025** (published)
Soumya Ghorai, Akhil Chambayil, Souvik Chattopadhyay

Conference Publications

A Dynamically Phase-Shifted PWM for Single-Stage Bipolar 3-Phase to 1-Phase Buck-Dab Converter
Botha Srinivas, **Soumya Ghorai**, Souvik Chattopadhyay **APEC 2027** (Digest Accepted)

Digital Dual Loop Control for Load Sharing of Parallel Resonant Converters
Sakshi Kumari, **Soumya Ghorai**, Souvik Chattopadhyay **PEDES 2026** (Digest Accepted)

Emulation of Volt-ampere Characteristics of Line Frequency Transformer by a Single-phase Single-stage AC-AC Solid State Transformer
Soumya Ghorai, Souvik Chattopadhyay **ECCE 2026**

A Universal On-board Charger for EVs with Minimum DC-link Capacitance
T Shibano, S Saha, **S Ghorai**, S Chattopadhyay, K Sakai, H Matsumori, T Kosaka **ECCE 2026**

Single-Phase Single-Stage Three-Port On-Board Charger with Minimum Hardware Components for EV/PHEV Application
K Ushida, K Nakamura, Y Hotta, S Saha, **S Ghorai**, S Chattopadhyay **ECCE 2025**

A New Three-phase Single-stage DAB Isolated Phase-modular PFC Topology with Sextuple Active Bridges
Soumya Ghorai, Souvik Chattopadhyay **ECCE Asia, 2025**

Dual Active Bridge-Based Isolated Single-phase Single-Stage DC-link AC/AC Converter with Minimum Charge Storage Requirement
Soumya Ghorai, Souvik Chattopadhyay **ECCE Asia, 2025**

A Highly Efficient Wide Battery Voltage Range Dual Active Bridge-Based Single-stage Bidirectional EV Onboard Charger Without Low-Frequency Harmonics in the Charging Current
Soumya Ghorai, Souvik Chattopadhyay **INTELEC, 2024**

PV assisted Fuzzy based EV charge scheduling for demand side energy management: a case study
Soumya Ghorai, Deepanjan Majumdar, Tushar Jash, Susanta Ray **CALCON, 2020**

Revisiting small signal stability analysis techniques in the context of stability study of a two-area multi-machine power system
Soumya Ghorai, Susanta Ray **CALCON, 2020**

Invention Dsiclosures

Soumya Ghorai, Souvik Chattopadhyay. *A Highly Efficient Active Power Decoupled Single-phase Single-stage EV Onboard Charging System With Ripple-less DC Battery Charging Current.*
Indian Patent (filed & early published App No: IN202431104682 A1)

Soumya Ghorai,Akhil Chambyil, Souvik Chattopadhyay. *A Modular Dual-Active Bridge (DAB)- based Three Phase AC-TO-DC Conversion System.*
Indian Patent (filed and early published App No: IN202531034478 A1)

Soumya Ghorai, Souvik Chattopadhyay. *A Novel Control & Modulation Technique for Single-stage Power Electronic Transformer.*
Indian Patent (filed & early pubished App No: IN202531034483 A1)

Research Experience

Ph.D. Project 1 (Execution Lead): Development of 50kW Single-stage DAB Isolated Modular DC Fast Charger
CPRI, India -funded Project PI: Dr. Souvik Chattopadhyay *August'23 - present*

- Conducted verification of a novel 3-phase modular single-stage AC-DC DAB topology in *PLECS* platform which consists of six active bridges with negligible DC-link capacitor requirement and enabled with direct line current shaping and minimum RMS current in the high-frequency link.

- Conducted PCB designing in *Altium*, 3D-design of mechanical assembly in *Solidworks*, hardware realization along with design and making the magnetics components, closed-loop-control implementation using Sprtan-6 FPGA board and *VHDL* coding in *XILINX*.
- Currently working towards testing the setup in full power, optimizing the modulation technique to enhance the soft-switching region while ensuring minimum high-frequency rms current during the entire grid cycle, and the ability to work under unbalanced grid voltage conditions.

Ph.D. Project 2 (Hardware Team IIT KGP lead): Simulation and Analysis of Universal EV On-Board Battery Charger Configurations with Wide-band gap-based power devices
AISIN CORPORATION, Japan - funded project PI: Souvik Chattopadhyay *Jan'24 - present*

- Conducted hardware realization of a 2.6 kW active power decoupled phase-shift modulated single-phase single-stage AC-DC DAB topology with minimum hardware components. A novel approach of instantaneous phase-shift modulation based on a predictive switching power modulator, is realized on FPGA platform and verified.

Other Projects 1 during Ph.D. (Co-developer of the Hardware): A figh-frequency isolated, soft-switched,single-stage,3-phase AC-DC bidirectional converter for battery storage application
ANRF,India - funded project PI: Souvik Chattopadhyay *March'26 - present*

- Implementing a dynamic phase-shifted PWM control for bipolar 3-phase to 1-phase Buck-DAB or Matrix DAB converter. The proposed technique doesn't not rely on look-up table or heavy online calculation, offers ease of implementation and immunity against grid disturbance or runtime parameter variation.

Other Projects 2 during Ph.D. (Co-developer of the Hardware): Grid Readiness for EV: Enablers and Technological developments
ANRF,India - funded project PI: Souvik Chattopadhyay *Oct'25 - present*

- Designed the testbed PCB to test CCS-2 EV charging protocol. Both high-level PLC via CP line and low-level PWM communication are being tested in the board with Spartan-3 XE3S400 FPGAs as mother controllers for EV and EVSE, whereas, Chargebyte-GmBH-made PLC modems are used to convert SPI signals to power line signals.

Other Projects 3 during Ph.D.: Testing of Railway Battery Charger (RBC) Power Converter
Scientific Power Corporation, Kolkata - funded project PI: Souvik Chattopadhyay *Jan'23 - present*

- Designed a customized control interface board for TI DSP TMS320F28379D for different relays, and display connection (UART communication), multiple PWM and analog sensing interfaces, SD card read and write option via interfacing with the STM 32 blue pill board via SPI communication, gate driver tripping input and output signals.

Past Projects 1: Soft-switched reduced switch high step-down converter
Research Scholar at Indian Institute of Technology Delhi *Aug'20 - March'22*
Supervisor: Mummadi Veerachary

- Designed and analyzed a new coupled-inductor-switched-capacitor-cell-based 48V-1V converter topology for VRM applications, which eliminates the leakage inductance caused by the voltage spike problem against the synchronous switch of bucking operation.

Past Projects 2: Off-grid battery-connected PV-fed UV-C LED driver for rural water purifier
Rashtriya Uchchatar Shiksha Abhiyan 2.0 Fellowship, Jadavpur University, Kolkata *June'19 - May'20*
PI: Kamalika Ghosh, Co-PI: Debashis Chatterjee, Dept. of Electrical Engineering

- Worked on a SEPIC converter interfacing battery and PV cells where duty is controlled via Perturb & Observe (P&O) MPPT method. Along with that a buck converter to interface the battery and UV-C leds.

Past Projects 3: Small Signal Stability Analysis of Power System
Research Intern at Indian Institute of Technology Delhi *May'18 - July'18*
Supervisor: Ashu Verma, Dept. of Energy Science and Engineering

- Analyzed small-signal stability of multi-bus multi-machine power system writing code in *Matlab*.

Academic Experiences

Relevant Coursework

	Offered by	SGPA	Sub. Code
Switch Mode Power Conversion	*S. Chattopadhyay, EE, IITKGP*	10/10	*EE60019*
Intro. to Electrical Machines and Drives	*G. Poddar, EE, IITKGP*	*10/10*	*EE61201*
Soft Switching Converters	*G. Poddar,EE, IITKGP*	*9/10*	EE60206
Statistical Methods	*B. Banerjee, Math, IITKGP*	*9/10*	MA60017
Selected Topics in Power Electronics	*M. Veerachary, EE, IITD*	*10/10*	ELL857
Advanced Topics in Power Electronics	*M. Veerachary, EE, IITD*	*10/10*	ELL856
Digital Control of Power Electronics	*S. Pramanick, EE, IITD*	*9/10*	ELL850
Optimal Control Theory	*I.N.Kar, EE, IITD*	*9/10*	ELL703

Teaching Assistantship

M.Tech. Power Electronics Lab	*Instructor : Prof. S. Chattopadhyay*	*IIT Kgp*
M.Tech. Machines Lab	*Instructor : Prof.G. Poddar*	*IIT Kgp*
B.Tech. 1st year Engineering Lab	*Instructor : Prof.T. Bhattacharya*	*IIT Kgp*
B.Tech. 4th year Power Apparatus and System Design Course	*Instructor : Prof.D. Kastha*	*IIT Kgp*
B.Tech. 1st year Introduction to Electrical Engineering	*Instructor : Prof.S.S. Nag*	*IIT Delhi*

References

Souvik Chattopadhyay souvik@ee.iitkgp.ac.in
Associate Professor,Indian Institute of Technology Kharagpur

Gautam Poddar gpoddar@ee.iitkgp.ac.in
Professor, Indian Institute of Technology Kharagpur

Subrata Saha subrata.saha@aisin.co.jp
Chief Technologist , Aisin corporation, Aichi, Japan

Debashis Chatterjee debashis.chatterjee@jadavpuruniversity.in
Professor, Jadavpur University